上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 1
Course Orientation

Haishan Yu



SJTU
饮 水 思 源 · 爱 国 荣 校

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Get to know each other, Expectations

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Learning Outcomes

1. **Describe** the course structure, learning activities, and assessment requirements.

2. **Identify** activities and information that may be relevant to estimating the emissions linked to a typical SJTU student.

3. **Formulate** initial questions about why carbon footprint estimates may differ.

01

Get to know each other

About me, about you, about expectations

Before the start

About me!

- Office hours:
  - By appointment
- Email: haishan.yu@sjtu.edu.cn

About you!

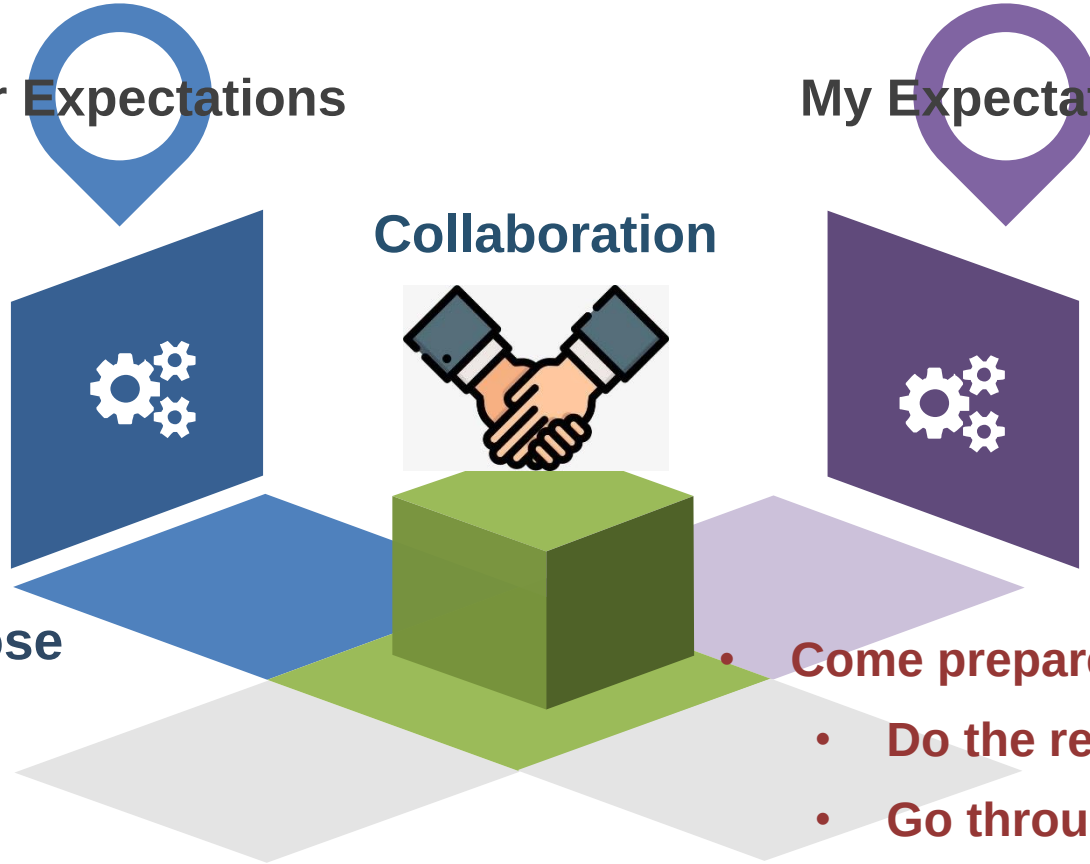


面对面建群：

0915



Expectations



Why motivate you to choose this course?

Stay in group with 2-3 students around you
&
Share your thoughts via "投稿 "
(3 min)

- Come prepared
  - Do the reading
  - Go through slides afterwards
- Work collaboratively (Project, Class and after-class discussion)
- Participate in class
  - Bring in ideas and interested questions
  - Being active in discussion
- Practice

02

Course Logistics

Course schedule, course assessment, learning outcomes

Course Information

- **Class time: Tuesdays** and **Thursdays**, 14:00-15:40
- **Venue:** SY-Upper Hall (上院) 403 (plz check the location before you come to class)
- **Course contents** (lecture notes, readings, assignments, announcements, etc. will be shared on **Canvas**)
- Most announcements will be made in **Wechat group** as well.
- **Collaborative learning**:
  - In-class discussion is mainly taken within groups.
  - After-class discussion is organized on Canvas-Discussions.

✓ Interactive, discussions, and exercises – Please ask questions!

✓ Practice and Critical thinking

Course Assessments

- Attendance and Participation – 10%
- In-class quiz – 20%
- Course project – 40%
- Final written Exam (restricted open-book) – 30%
  - Exact date to be decided

Course Project Overview

- **Course Project: Personal Carbon Footprint Calculator**
  Group work: 4–5 students, 40% of the course grade
- Your challenge is to work with AI to design, test and progressively improve a web-based Personal Carbon Footprint Calculator for a shared virtual SJTU student. No prior programming experience is required.
- You will:
  1. **Understand your client** and identify what the calculator should help them do.
  2. **Create Prototype 0** with AI assistance, based on your group's initial ideas.
  3. **Test and revisit** your calculator at several checkpoints as the course progresses.
  4. **Improve the product** using carbon accounting concepts, data, evidence and user feedback.
  5. **Present and demonstrate** your final calculator at the end of the course.
- AI can help you build the calculator. Your group remains responsible for deciding what should be built—and whether the result is fit for purpose.

Course Project - How Your Group Will Work

Everyone Contributes

Each group member is expected to make a meaningful contribution to one or more of the following—and to understand the group's overall work:

- Client and requirement discussions
- Interaction with AI and prompt development
- Calculator and interface development
- Testing and result checking
- Product revision and documentation
- Data collection and source evaluation
- Final demonstration and presentation

Keep a Record of Your Journey

Throughout the project, your group will document:

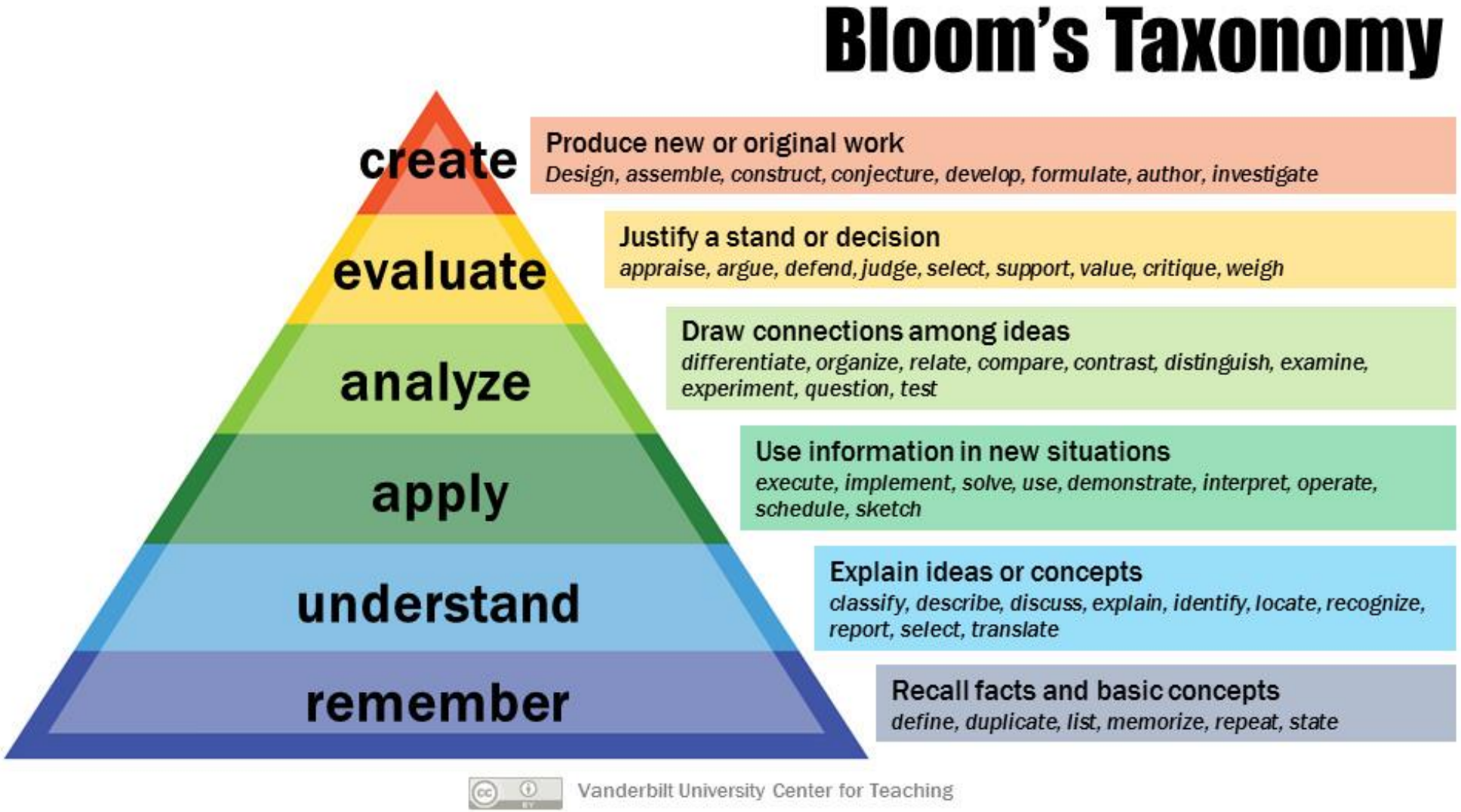
- Your evolving client profile and assumptions
- Prompts and relevant AI interactions
- Important design and accounting decisions
- Tests, problems and unexpected results
- What you revised, why you revised it, and what evidence informed the change
- Contributions and discussion memos from group members

Course Project – Assessment (40%)

Component	Weight	What Will Be Assessed
Development and Iteration Portfolio	15%	Client profiles, prompts, testing records, successive versions, discussion memos, and revision rationales
Final Web App	15%	Accounting quality, client fit, functionality, usability, transparency, and interpretation of results
Final Presentation and Q&A	5%	Product demonstration, explanation of key decisions and revisions, and responses to questions
Individual Contribution and Reflection	5%	Individual contribution, reflection, peer feedback, and understanding of the group's work

Module I	Lecture 1	Week 1	9/15	Course Orientation	Lecture + Discussion
	Lecture 2	Week 1	9/17	What is Carbon Accounting and Why Do It	Lecture + Discussion
	Lecture 3	Week 2	9/22	How Carbon Accounting Works	Lecture + Discussion
	Lecture 4	Week 2	9/24	Attributional vs Consequential Accounting	Lecture + Discussion
Module II	Lecture 5	Week 3	9/29	National Level Carbon Accounting	Lecture + Discussion
	Lecture 6	Week 4	10/8	Communtity Level Carbon Accounting	Lecture + Discussion
	Lecture 7	Week 5	10/13	Orgaizational Level Carbon Accouting I	Lecture + Discussion
	Lecture 8	Week 5	10/15	Orgaizational Level Carbon Accouting II	Lecture + Discussion
	Lecture 9	Week 6	10/20	Product Level Carbon Accounting I	Lecture + Discussion
	Lecture 10	Week 6	10/22	Product Level Carbon Accounting II	Lecture + Discussion
	Lecture 11	Week 7	10/27	Project and Policy Level Carbon Accounting	Lecture + Discussion
	Lecture 12	Week 7	10/29	Financial Carbon Accounting	Lecture + Discussion
Module III	Lecture 13	Week 8	11/3	Reporting and Disclosure I	Lecture + Discussion
	Lecture 14	Week 8	11/5	Reporting and Disclosure II	Lecture + Discussion
Module IV	Lecture 15	Week 9	11/10	Project Presentations	Presentation
	Exam	Week 9	TBA		Examination

Learning Outcomes



Learning Outcomes (Knowledge)

- **Foundations:** Explain the purposes, key concepts, and basic quantification logic of carbon accounting.
- **Carbon accounting structure**: Analyze how purpose, unit of analysis, boundary, attribution, data, methods, and intended use shape accounting results.
- **Applications:** Distinguish carbon accounting across national, city, organizational, product, project and policy, and financial contexts.
- **Quality and credibility:** Argue how data quality, assumptions, uncertainty, and comparability affect the credibility and use of results.
- **Responsibility and action:** Argue how accounting choices influence the attribution of responsibility and mitigation actions.

Learning Outcomes (Skills)

- **Problem framing:** Interpret the purpose, unit of analysis, boundary, attribution rules, data needs, and intended use of an accounting task.
- **Quantification:** Calculate simplified inventories and carbon footprints, documenting key data sources and assumptions.
- **Critical judgment:** Compare accounting approaches and evaluate their credibility, limitations, and fitness for purpose.
- **AI-assisted inquiry:** Identify knowledge gaps, assess evidence, and use generative AI to design, test and improve models while retaining responsibility for final decisions.
- **Teamwork:** Coordinate tasks and negotiate key accounting decisions in diverse teams.
- **Communication:** Explain accounting choices, results, uncertainty, and limitations clearly to different audiences..

Any Questions?

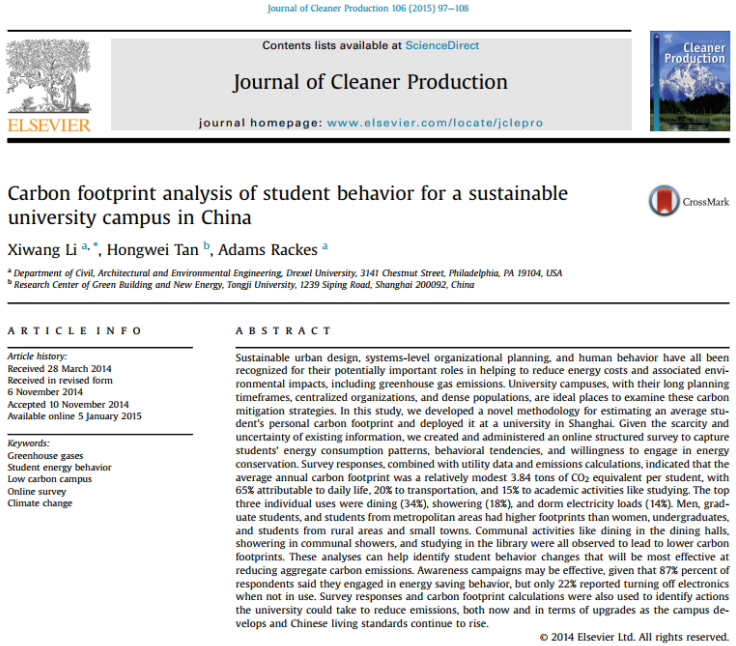
03

A first look at Carbon Accounting

Think about student emissions

Can These Estimates Be Compared?

Tongji University:
3.84 tCO_2e/student/year



Journal of Cleaner Production 106 (2015) 97–108

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Carbon footprint analysis of student behavior for a sustainable university campus in China

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ARTICLE INFO

Article history:
Received 28 March 2014
Received in revised form
6 November 2014
Accepted 10 November 2014
Available online 5 January 2015

Keywords:
Greenhouse gases
Student energy behavior
Low carbon campus
Online survey
Climate change

ABSTRACT

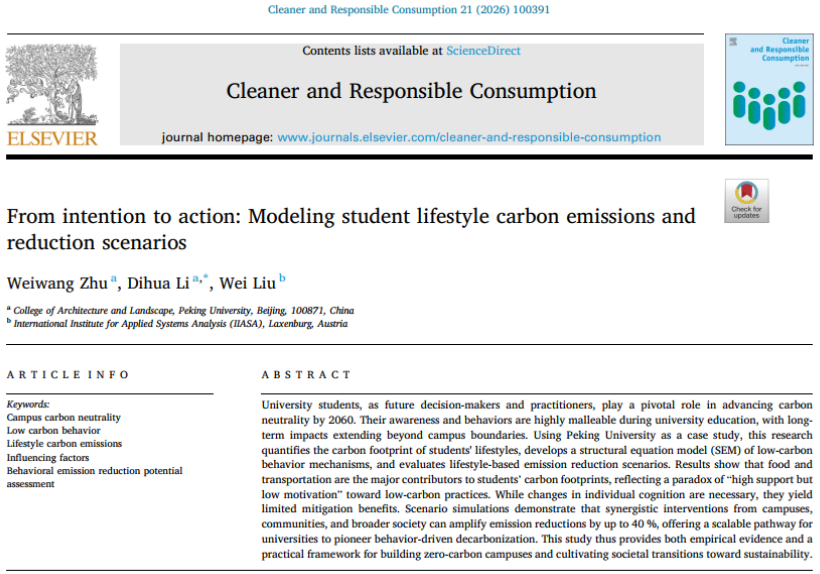
Sustainable urban design, systems-level organizational planning, and human behavior have all been recognized for their potentially important roles in helping to reduce energy costs and associated environmental impacts, including greenhouse gas emissions. University campuses, with their long planning timeframes, centralized organizations, and dense populations, are ideal places to examine these carbon mitigation strategies. In this study, we developed a novel methodology for estimating an average student's personal carbon footprint and deployed it at a university in Shanghai. Given the scarcity and uncertainty of existing information, we created and administered an online structured survey to capture students' energy consumption patterns, behavioral tendencies, and willingness to engage in energy conservation. Survey responses, combined with utility data and emissions calculations, indicated that the average annual carbon footprint was a relatively modest 3.84 tons of CO_2 equivalent per student, with 65% attributable to daily life, 20% to transportation, and 15% to academic activities like studying. The top three individual uses were dining (34%), showering (18%), and dorm electricity loads (14%). Men, graduate students, and students from metropolitan areas had higher footprints than women, undergraduates, and students from rural areas and small towns. Communal activities like dining in the dining halls, showering in communal showers, and studying in the library were all observed to lead to lower carbon footprints. These analyses can help identify student behavior changes that will be most effective at reducing aggregate carbon emissions. Awareness campaigns may be effective, given that 87% percent of respondents said they engaged in energy saving behavior, but only 22% reported turning off electronics when not in use. Survey responses and carbon footprint calculations were also used to identify actions the university could take to reduce emissions, both now and in terms of upgrades as the campus develops and Chinese living standards continue to rise.

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http://dx.doi.org/10.1016/j.jclepro.2014.11.084



Peking University:
2.54 tCO_2e/student/year



Cleaner and Responsible Consumption 21 (2026) 100391

Contents lists available at ScienceDirect

Cleaner and Responsible Consumption

journal homepage: www.journals.elsevier.com/cleaner-and-responsible-consumption

From intention to action: Modeling student lifestyle carbon emissions and reduction scenarios

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ARTICLE INFO

Keywords:
Campus carbon neutrality
Low carbon behavior
Lifestyle carbon emissions
Influencing factors
Behavioral emission reduction potential assessment

ABSTRACT

University students, as future decision-makers and practitioners, play a pivotal role in advancing carbon neutrality by 2060. Their awareness and behaviors are highly malleable during university education, with long-term impacts extending beyond campus boundaries. Using Peking University as a case study, this research quantifies the carbon footprint of students' lifestyles, develops a structural equation model (SEM) of low-carbon behavior mechanisms, and evaluates lifestyle-based emission reduction scenarios. Results show that food and transportation are the major contributors to students' carbon footprints, reflecting a paradox of "high support but low motivation" toward low-carbon practices. While changes in individual cognition are necessary, they yield limited mitigation benefits. Scenario simulations demonstrate that synergistic interventions from campuses, communities, and broader society can amplify emission reductions by up to 40 %, offering a scalable pathway for universities to pioneer behavior-driven decarbonization. This study thus provides both empirical evidence and a practical framework for building zero-carbon campuses and cultivating societal transitions toward sustainability.

https://doi.org/10.1016/j.clrc.2026.100391

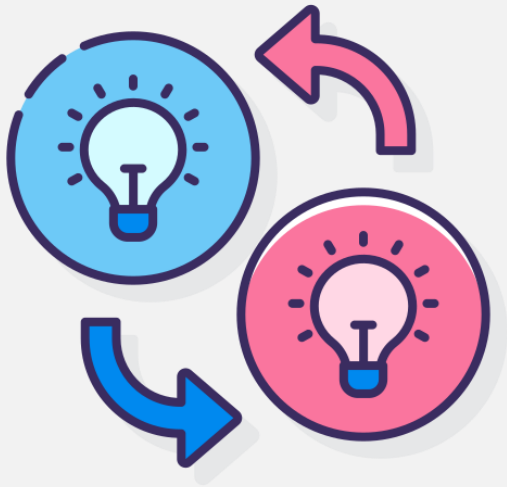
Can we conclude that PKU students have a lower carbon footprint?

Prelude: Thinking about Student Emissions

Think — 4 min



Pair — 6 min



Share — 5 min



Imagine a simple tool that estimates the greenhouse gas emissions linked to the activities of a typical SJTU student.

Step 1 – Think: Your Initial Ideas

Work individually (See Hand-out).

1. What might the estimates help an SJTU student understand?

2. Which activities might be relevant? List up to five.

3. Choose one activity. What would we need to know about the activity.

Time: 4 minutes



Step 2 – Compare and Combine (6 min)



Compare your individual ideas. AND

Complete one discussion sheet (see hand-out)

1. Agree on one possible use of the estimate.
2. Select up to four activities. AND write what we would need to know about each activity.
3. Identify one activity you remain unsure about.

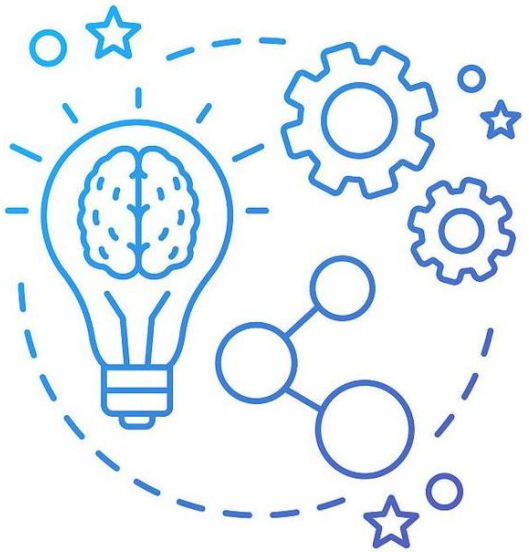
Time: 6 minutes

Step 3 – Share * (5 min)

Each pair reports only one item:

Which activity was hardest to include or leave out?

Be ready to explain.



Carbon accounting

Carbon Accounting ?

Carbon?

Carbon Peak

Carbon Neutrality

Carbon

Carbon

Carbon

Carbon

Accounting?

Accounting:
https://www.merriam-webster.com/dictionary/accounting：
the system of **recording** and **summarizing** business and financial transactions and **analyzing**, **verifying**, and **reporting** the results
also：the **principles** and **procedures** of this system

核算（会计核算）：
以货币为**计量单位**，运用**专门的**会计方法，对生产经营活动或预算执行过程及其结果进行**连续**、**系统**、**全面**的**记录**、**计算**、**分析**，**定期编制**并提供财务会计**报告**和其他会计资料，为经营决策和宏观管理**提供依据**的一项会计活动。

03

Wrap-up and Next Time

Wrap-up

Today

- We reviewed the course structure, learning activities, and assessment.
- We considered what a student emissions estimate might help us understand.
- Different pairs identified different activities, information needs, and uncertainties.

Next time

- What Is Carbon Accounting, and Why Do It?
- Kick-off of Course Project

Optional discussion task on Canvas

What explains the difference between PKU and Tongji student carbon footprint?

Use AI to explore why the PKU and Tongji estimates differ.

Post (see Canvas-Discussions):

- Your exact question to AI
- AI's answer in no more than three bullet points
- One claim checked against either paper
- One better follow-up question

Optional: A thoughtful post can count as evidence of course participation.

AI output is a starting point, not evidence by itself.

Suggested Readings for Lecture 2

- **Essential reading:**
  - Ascui, F. and Lovell, H. (2011) “As frames collide: Making sense of carbon accounting”, AAAJ 24 (8): 978-999) [Read Sections 2, 4.1 and 4.2, roughly 4 pages; If you wish, feel free to read through the whole of section 4]
- **Optional reading**
  - Li, X., Tan, H. and Rackes, A. (2015) “Carbon footprint analysis of student behavior for a sustainable university campus in China”, Journal of Cleaner Production 106: 97–108.
  - Zhu, W., Li, D. and Liu, W. (2026) “From intention to action: Modeling student lifestyle carbon emissions and reduction scenarios”, Cleaner and Responsible Consumption 21: 100391.

  (You do not need to read both papers in full; only read the sections that help you prepare for the Canvas discussion task “Two Estimates, One Question”.)
- **Further reading:**
  - https://ourworldindata.org/ghg-emissions-by-sector





See you next time